SPIRONOLACTONE

Hyperkalaemia
Gynaecomastia

CHEMICAL NAME S-[(7R,8R,9S,10R,13S,14S,17R)-10,13-dimethyl-3,5′-dioxospiro[2,6,7,8,9,11,12,14,15,16-decahydro-1H-cyclopenta[a]phenanthrene-17,2′-oxolane]-7-yl] ethanethioate

CLASS Potassium-sparing diuretic; mineralocorticoid receptor antagonist (MRA)

TRADE NAMES Aldactone · Spiractin · Aldactide (with HCTZ)

THERAPEUTIC USES / INDICATIONS

- Heart failure with reduced EF (↓ mortality, RALES).
- Ascites in liver cirrhosis (first-line diuretic).
- Primary hyperaldosteronism (Conn's syndrome); resistant hypertension.
- Nephrotic syndrome oedema; hirsutism, acne in PCOS (off-label).

ADMINISTRATION

- **PO** once daily with food (↑ absorption).
- Effect builds over several days.

DOSAGES

- **Heart failure:** 25 mg once daily (12.5 mg if K^+ or renal concern); max 50 mg.
- **Ascites:** 100 mg daily, increase to 400 mg (with furosemide in 100:40 ratio).
- **Resistant hypertension:** 25–50 mg daily.
- **Hyperaldosteronism:** 100–400 mg daily.

PHARMACOKINETICS

- **Onset:** gradual — 2–3 days for full diuresis.
- Metabolised in liver to active canrenone and other metabolites.
- Excreted in urine and bile. **t½:** 1.4 h (active metabolites 13–24 h).

NURSING MANAGEMENT

- Check K^+ and creatinine at baseline, 3 days, 1 week, then monthly for 3 months.
- Hold if K^+ >5.5 and report.
- Daily weight; abdominal girth in ascites.
- Monitor for gynaecomastia.
- Review other K^+-raising drugs.

PHARMACODYNAMICS

- Competitively blocks **aldosterone receptors** in distal tubule/collecting duct → ↑ Na^+ and water excretion, K^+ retention.
- Reduces myocardial fibrosis in HF.
- Anti-androgenic effects.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Gynaecomastia, breast tenderness (men).
- Menstrual irregularities, reduced libido.
- GI upset, dizziness.

ADVERSE EFFECTS (SERIOUS)

- Hyperkalaemia (dangerous with ACEi/ARB, renal impairment).
- Hyponatraemia.
- Acute kidney injury.
- Metabolic acidosis.

CONTRAINDICATIONS & CAUTIONS

- Hyperkalaemia (K^+ >5.0); Addison's disease.
- Anuria, severe renal impairment (eGFR <30).
- Use with eplerenone or K^+ supplements.
- Pregnancy (anti-androgenic).

KEY INTERACTIONS

- **ACE inhibitors, ARBs, trimethoprim, NSAIDs, K^+ supplements, salt substitutes:** hyperkalaemia.
- **Digoxin:** ↑ digoxin levels.
- **Lithium:** ↑ levels.

PATIENT EDUCATION

- Avoid potassium salt substitutes and high-potassium supplements.
- Report muscle weakness, palpitations, breast swelling.
- Stop and seek advice if vomiting/diarrhoea (risk of high K^+ and kidney injury).
- Blood tests are essential.

KEY POINTS

- Aldosterone antagonist; K^+-sparing.
- Hyperkalaemia, gynaecomastia.
- HFrEF mortality benefit; first-line for ascites.
- Check K^+ regularly, esp. with ACEi/ARB.